Application for CCRI Summer Internship Program (Online)

1 June – 31 August 2026

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To

Prof. Ching-Hsien Hsu
Chair Professor, CSIE Department & Director, Big Data Research Center
Asia University, Taiwan

Cc: Prof. Min-Shiang Hwang, Prof. Whai-En Chen, Prof. Chun-Yuan Lin, Prof. Charles C.N. Wang
Faculty Members, International Center for AI and Cyber Security Research and Innovations (CCRI)

Dear Prof. Ching-Hsien Hsu and Esteemed CCRI Faculty Members,

I am writing to apply for the CCRI Summer Internship Program (Online) 2026. As a third-year Computer Science student at IISER Bhopal deeply engaged in quantum information, quantum algorithms, and cybersecurity, I am excited to contribute to research at the intersection of AI, machine learning, and quantum-secure systems.

1. Introduction and Background

Quantum Key Distribution (QKD) offers information-theoretic security based on quantum mechanics. However, real-world deployments face significant challenges from implementation vulnerabilities, side-channel attacks, and sophisticated adversarial interference in both quantum and classical post-processing layers. My prior work on anomaly detection for QKD systems, combined with expertise in quantum-inspired algorithms, stochastic simulation, and optimization, positions me to develop advanced AI-driven solutions for quantum cybersecurity.

This proposal outlines a concrete 3-month research project that builds directly on my experience while aligning with CCRI's mission in AI and Cybersecurity.

2. Proposed Project: Hybrid AI-Quantum Anomaly Detection Framework for Secure Quantum Communication Networks

Project Title: Intelligent Threat Detection and Mitigation in Quantum Key Distribution Systems using Quantum-Inspired Machine Learning and Optimization Techniques

Objectives

- Develop a hybrid quantum-classical anomaly detection system capable of identifying both known and novel attacks on QKD protocols with high accuracy and low latency.
- Integrate quantum-inspired techniques (amplitude encoding and multi-observable uncertainty quantification) with modern AI methods for improved detection in noisy environments.
- Leverage structured optimization and stochastic simulation methods to enhance model robustness and scalability.

- Evaluate the framework under realistic noise models, side-channel attacks, and high-throughput scenarios.

Detailed Methodology

1. **Data Generation and Simulation** Extend my previous QKD anomaly detection work by building realistic simulation environments incorporating quantum channel noise, photon loss, and various attack vectors (photon-number-splitting, detector blinding, Trojan-horse attacks, etc.).

2. **Quantum-Inspired Feature Engineering** Apply amplitude encoding in Hilbert space and covariance-aware acquisition functions (from my NQComp 2026 materials discovery project) to model cross-parameter correlations in QKD signals.

3. **Advanced Modeling** - Combine Bayesian probabilistic estimators (from my sktime/skpro work) with quantum kernel methods. - Use optimization techniques from my JuliaSmoothOptimizers (JSO) internship to tune model parameters efficiently. - Incorporate insights from ongoing quantum convolutional codes and entanglement distillation research at UMass Amherst (NSF CQN) for error-aware detection.

4. **Implementation & Optimization** Implement low-latency pipelines suitable for edge deployment, drawing from my GeoAI4Cities Jetson Nano experience. Explore GPU/parallel acceleration where applicable.

5. **Evaluation** Metrics: Detection accuracy, false positive rate, latency, QBER impact, and scalability. Benchmark against classical ML baselines and my previous 95.2% accuracy framework.

Expected Outcomes

- A robust, open-source hybrid anomaly detection module for QKD systems.
- Comprehensive performance analysis and potential contribution to a research paper.
- Practical recommendations for integration into real quantum communication infrastructures.

3. Relevance to My Background

This project directly extends my hands-on experience:

- **QNu Labs (2025)**: Designed QKD anomaly detection framework (95.2% accuracy, 2.1% FPR, sub-50ms latency).
- **Quantum-Inspired Active Learning (NQComp 2026)**: Developed amplitude encoding and uncertainty aggregation techniques.
- **Lattice Simulations (Univ. Cambridge)**: Strong foundation in stochastic methods (HMC, Metropolis-Hastings).
- **JSO Internship (Polytechnique Montréal)**: Structured optimization and Julia ecosystem contributions.
- **UMass Amherst (NSF CQN)**: Quantum convolutional codes and quantum repeater simulations.
- Systems experience: Linux kernel contributions and edge AI deployment.

4. Timeline (1 June – 31 August 2026)

Period	Major Activities
Week 1–2	Literature review, project setup, simulation environment
Week 3–5	Data generation, quantum-inspired feature engineering, baseline models
Week 6–8	Implementation of hybrid detection models and optimization
Week 9–11	Integration of error correction insights, extensive testing & attack simulations
Week 12	Final evaluation, documentation, report, and presentation

I am fully committed to the program and will dedicate 35–40 hours per week. I will maintain regular communication with mentors via email, meetings, and progress reports.

5. Conclusion

I am enthusiastic about the opportunity to work with CCRI's distinguished faculty and contribute to cutting-edge research in AI-driven quantum cybersecurity. This internship represents an ideal platform to advance my skills while delivering impactful results for secure quantum networks.

Thank you for considering my application. I am available for any further discussion.

Sincerely,

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